

Git Hands-On 4 – Merge Conflict Resolution

1. Create a new branch GitWork and add a file hello.xml

```
JONITH J V@LAPTOP-93Q0MAID MINGW64 ~
$ git status
fatal: not a git repository (or any of the parent directories): .git

JONITH J V@LAPTOP-93Q0MAID MINGW64 ~
$ git checkout -b GitWork
touch hello.xml
fatal: not a git repository (or any of the parent directories): .git

JONITH J V@LAPTOP-93Q0MAID MINGW64 ~
$ cd ~/GitDemo

JONITH J V@LAPTOP-93Q0MAID MINGW64 ~/GitDemo (main)
$ git status
On branch main
Your branch is ahead of 'origin/master' by 4 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean

JONITH J V@LAPTOP-93Q0MAID MINGW64 ~/GitDemo (main)
$ git checkout -b GitWork
touch hello.xml
Switched to a new branch 'GitWork'

JONITH J V@LAPTOP-93Q0MAID MINGW64 ~/GitDemo (GitWork)
$ echo "<message>Hello from GitWork Branch</message>" > hello.xml
git status
On branch GitWork
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hello.xml

nothing added to commit but untracked files present (use "git add" to track)
```

Step 2: Commit the changes to master

```
JONITH J V@LAPTOP-93Q0MAID MINGW64 ~/GitDemo (GitWork)
$ git add hello.xml
git commit -m "Added hello.xml in GitWork branch"
warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next time Git touches it
diff --git a/.gitignore b/.gitignore
new file mode 100644
index 0000000..35bf57b
--- /dev/null
+++ b/.gitignore
@@ -0,0 +1,2 @@
+*.log
+log/
diff --git a/README.md b/README.md
new file mode 100644
index 0000000..11df79a
--- /dev/null
+++ b/README.md
@@ -0,0 +1 @@
+# GitDemo
\ No newline at end of file
diff --git a/branch.txt b/branch.txt
new file mode 100644
index 0000000..d72aca6
--- /dev/null
+++ b/branch.txt
@@ -0,0 +1 @@
+This file was created in GitNewBranch.
:...skipping...
diff --git a/.gitignore b/.gitignore
new file mode 100644
index 0000000..35bf57b
--- /dev/null
+++ b/.gitignore
@@ -0,0 +1,2 @@
+*.log
+log/
```

Step 3 : Merge the GitWork branch into master

```
JONITH J V@LAPTOP-93Q0MAID MINGW64 ~  
$ cd ~/GitDemo  
  
JONITH J V@LAPTOP-93Q0MAID MINGW64 ~/GitDemo (master)  
$ git merge GitWork  
Auto-merging hello.xml  
CONFLICT (add/add): Merge conflict in hello.xml  
Automatic merge failed; fix conflicts and then commit the result.
```

```
$ cat hello.xml  
<<<<<< HEAD  
<message>Hello from Main Branch</message>  
=====  
<message>Hello from GitWork Branch</message>  
>>>>>> GitWork
```

```
$ git add hello.xml
```

```
$ cat hello.xml  
<message>Hello from Main Branch</message>  
<message>Hello from GitWork Branch</message>
```

```
$ git commit -m "Resolved merge conflict in hello.xml"  
[master b6b0c62] Resolved merge conflict in hello.xml
```

Step 4: Commit the .gitignore file and List all available branches

```
JONITH J V@LAPTOP-93Q0MAID MINGW64 ~/GitDemo (master)  
$ git add .gitignore  
git commit -m "Added backup files to gitignore"  
On branch master  
Your branch is ahead of 'origin/master' by 7 commits.  
  (use "git push" to publish your local commits)  
  
nothing to commit, working tree clean  
  
JONITH J V@LAPTOP-93Q0MAID MINGW64 ~/GitDemo (master)  
$ git branch  
  GitWork  
  main  
* master
```

Step 5: Deletion of Branch

```
JONITH J V@LAPTOP-93Q0MAID MINGW64 ~/GitDemo (master)  
$ git branch -d GitWork  
Deleted branch GitWork (was 36fad21).
```

Step 6 : Viewed the Commit History

```
JONITH J V@LAPTOP-93Q0MAID MINGW64 ~/GitDemo (master)  
$ git log --oneline --graph --decorate  
* 19d1458 (HEAD -> master) Resolved merge conflict  
| \  
| * 36fad21 Added hello.xml in GitWork branch  
| * bf5af3e (origin/main, main) Added branch.txt in GitNewBranch  
| * ec6d867 Successfully configured .gitignore for log files and folders  
| * 87c25e5 Merge remote changes  
| \  
| | * 9dfb736 Initial commit  
| | * eba6816 Added hello.xml in master  
| | \  
| | * 97f3d5a (origin/master) Initial Commit
```

Step 7 : Gitpush

```
JONITH J V@LAPTOP-93Q0MAID MINGW64 ~/GitDemo (master)
$ git push origin master
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 8 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (9/9), 963 bytes | 963.00 KiB/s, done.
Total 9 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To https://github.com/jonithjo/GitDemo.git
  97f3d5a..19d1458  master -> master
```

Github Repository

